Use case	Registration	
Release (id)	UC1, v0.1, 19/04/2026	
Context of Use	User authentication and account management.	
Scope	Let an anonymous user create a new account.	
Level	Primary task	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to create an account
	System	Manage the user interface, the database and their interactions
Preconditions	The user must be logged out.	
Minimal Guarantees	No duplicated or unsecure account is created.	
Success Guarantees	The anonymous user successfully created an account and is logged in.	
Trigger	The anonymous user selects “register”.	
Description	**Step**	**Action**
	1	The anonymous user selects the functionality “register”.
	2	The system asks the user to insert username and email.
	3	The user inserts the required data.
	4	The system checks whether an account with at least one between username and email already exists.
	5	The system asks the user to insert the password twice.
	6	The user inserts the required data.

	7	The system checks whether the password satisfies the security requirements and if the two passwords coincide.
	8	The system asks the user's consent for personal data processing.
	9	The anonymous user accepts the personal data processing.
	10	The system saves the data in the account archive.
	11	The system notifies the user of successful registration with an email.
	12	The system switches to the authenticated user functionalities.
Extension	**Step**	**Other action**
	4a	Invalid username or email: UC1.1
	7a	Invalid password: UC1.2
Technology & data Variations	None	

Alternative flow	Registration: Invalid username or email	
Release (id)	UC1.1, v0.2, 25/04/2026	
Context of Use	User authentication and account management.	
Scope	The system alerts the anonymous user that they have inserted an invalid username or email.	
Level	Subfunction	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to create an account
	System	Manage the user interface, the database and their interactions
Preconditions	The user inserted an already existing username or email.	
Minimal Guarantees	No duplicated account is created.	
Success Guarantees	None	
Trigger	UC1	
Description	**Step**	**Action**
	1	The system alerts the user that has inserted an invalid username or email.
	2	The sequence of events starts at step 2 of the main flow (UC1).
Technology & data Variations	None	

Alternative flow	Registration: Invalid password	
Release (id)	UC1.2, v0.2, 25/04/2026	
Context of Use	User authentication and account management.	
Scope	The system alerts the anonymous user that they have inserted an invalid password.	
Level	Subfunction	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to create an account
	System	Manage the user interface, the database and their interactions
Preconditions	The user inserted a password that does not satisfy the security requirements or the two passwords do not coincide.	
Minimal Guarantees	No unsecure account is created.	
Success Guarantees	None	
Trigger	UC1	
Description (Main Success Scenario)	**Step**	**Action**
	1	The system alerts the user that has inserted an invalid password, specifying the problem.
	2	The sequence of events starts at step 5 of the main flow (UC1).
Technology & data Variations	None	

Use case	Account deletion	
Release (id)	UC2, v0.1, 19/04/2026	
Context of Use	Account management.	
Scope	Let an authenticated user delete her account.	
Level	Primary task	
Primary actor	Passenger or train company	
Stakeholder & interest	**Stakeholder**	**Interest**
	Passenger	Wants to delete her account
	Train company	Wants to delete its account
	System	Manage the user interface, the database and their interactions
Preconditions	The user must be logged in.	
Minimal Guarantees	None	
Success Guarantees	The user successfully deleted her account and is logged out.	
Trigger	The user selects “delete account”.	
Description	**Step**	**Action**
	1	The anonymous user selects the functionality “delete account”.
	2	The system removes the data in the account archive.
	3	The system notifies the user of successful deletion with an email.
	4	The system switches to the unauthenticated user functionalities.
Extension	**Step**	**Other action**
	-	-

Technology & data Variations	None

Use case	Account data view	
Release (id)	UC3, v0.2, 25/04/2026	
Context of Use	Account management.	
Scope	Let an authenticated user view her account's data.	
Level	Primary task	
Primary actor	Authenticated user (passenger, train company or network manager)	
Stakeholder & interest	**Stakeholder**	**Interest**
	Authenticated user	Wants to view her account's data
	System	Manage the user interface, the database and their interactions
Preconditions	The user must be logged in.	
Minimal Guarantees	None	
Success Guarantees	The user successfully views her account's data.	
Trigger	The anonymous user selects "view account's data".	
Description	**Step**	**Action**
	1	The anonymous user selects the functionality "view account's data".
	2	The system shows the data saved in the account archive, that is username and email.
Extension	**Step**	**Other action**
	-	-
Technology & data Variations	None	

Use case	Account data change	
Release (id)	UC4, v0.2, 25/04/2026	
Context of Use	Account management.	
Scope	Let an authenticated user change her account's data.	
Level	Primary task	
Primary actor	Authenticated user (passenger, train company or network manager)	
Stakeholder & interest	**Stakeholder**	**Interest**
	Authenticated user	Wants to change her account's data
	System	Manage the user interface, the database and their interactions
Preconditions	The user must be logged in.	
Minimal Guarantees	No duplicated or unsecure account is created with a modification.	
Success Guarantees	The authenticated user successfully changed her account's data.	
Trigger	The authenticated user selects "change data".	
Description	**Step**	**Action**
	1	The authenticated user selects the functionality "change data".
	2	The system asks the user which data to modify.
	3	If the user selects username
	3.1	The system asks the user to insert the new username.
	3.2	The user inserts the new username.
	3.3	The system checks whether an account with the same username already exists.

	4	If the user selects email
	4.1	The system asks the user to insert the new email.
	4.2	The user inserts the new email.
	4.3	The system checks whether an account with the same email already exists.
	5	If the user selects password
	5.1	The system asks the user to insert the password twice.
	5.2	The user inserts the required data.
	5.3	The system checks whether the password satisfies the security requirements and if the two passwords coincide.
	6	The system saves the data in the account archive.
	7	The system notifies the user of successful data change with an email.
Extension	**Step**	**Other action**
	3.3a	Invalid username or email: UC4.1
	4.3a	Invalid email: UC4.2
	5.3a	Invalid password: UC4.3
Technology & data Variations	None	

Alternative flow	Account data change: Invalid username	
Release (id)	UC4.1, v0.2, 25/04/2026	
Context of Use	Account management.	
Scope	The system alerts the authenticated user that they have inserted an invalid username.	
Level	Subfunction	
Primary actor	Authenticated user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Authenticated user	Wants to change username
	System	Manage the user interface, the database and their interactions
Preconditions	The user inserted an already existing username.	
Minimal Guarantees	No duplicated account is created.	
Success Guarantees	None	
Trigger	UC4	
Description	**Step**	**Action**
	1	The system alerts the user that has inserted an invalid username.
	2	The sequence of events starts at step 3.1 of the main flow (UC4).
Technology & data Variations	None	

Alternative flow	Account data change: Invalid email	
Release (id)	UC4.2, v0.2, 25/04/2026	
Context of Use	Account management.	
Scope	The system alerts the authenticated user that they have inserted an invalid email.	
Level	Subfunction	
Primary actor	Authenticated user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Authenticated user	Wants to change email
	System	Manage the user interface, the database and their interactions
Preconditions	The user inserted an already existing email.	
Minimal Guarantees	No duplicated account is created.	
Success Guarantees	None	
Trigger	UC4	
Description	**Step**	**Action**
	1	The system alerts the user that has inserted an invalid email.
	2	The sequence of events starts at step 4.1 of the main flow (UC4).
Technology & data Variations	None	

Alternative flow	Registration: Invalid password	
Release (id)	UC4.3, v0.2, 25/04/2026	
Context of Use	Account management.	
Scope	The system alerts the authenticated user that they have inserted an invalid password.	
Level	Subfunction	
Primary actor	Authenticated user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Authenticated user	Wants to change password
	System	Manage the user interface, the database and their interactions
Preconditions	The user inserted a password that does not satisfy the security requirements or the two passwords do not coincide.	
Minimal Guarantees	No unsecure account is created.	
Success Guarantees	None	
Trigger	UC4	
Description (Main Success Scenario)	**Step**	**Action**
	1	The system alerts the user that has inserted an invalid password, specifying the problem.
	2	The sequence of events starts at step 5.1 of the main flow (UC4).
Technology & data Variations	None	

Use case	Reset password	
Release (id)	UC5, v0.1, 25/04/2026	
Context of Use	Account management.	
Scope	Let an anonymous user reset the password.	
Level	Primary task	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to reset the password
	System	Manage the user interface, the database and their interactions
Preconditions	The user must be logged out.	
Minimal Guarantees	No unsecure account is created with a modification. No user can change the password of another user.	
Success Guarantees	The anonymous user successfully changed the password.	
Trigger	The anonymous user selects “forgot password”.	
Description	**Step**	**Action**
	1	The anonymous user selects the functionality “forgot password”.
	2	The system asks the user to insert the email associated to her account.
	3	The user inserts the email.
	4	The system checks if the email is associated to an existing account.
	5	The system creates a random identifier.
	6	The system sends the identifier to the email.
	7	The system asks the user to insert the identifier it sent.

	8	The user inserts the identifier.
	9	The system checks whether the identifier is correct.
	10	The system asks the user to insert the password twice.
	11	The user inserts the required data.
	12	The system checks whether the password satisfies the security requirements and if the two passwords coincide.
	13	The system saves the data in the account archive.
	14	The system notifies the user of successful password change with an email.
Extension	**Step**	**Other action**
	4a	Invalid email: UC5.1
	9a	Invalid identifier: UC5.2
	12a	Invalid password: UC5.3
Technology & data Variations	None	

Alternative flow	Reset password: Invalid email	
Release (id)	UC5.1, v0.1, 25/04/2026	
Context of Use	Account management.	
Scope	The system alerts the authenticated user that they have inserted an invalid email.	
Level	Subfunction	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to reset password
	System	Manage the user interface, the database and their interactions
Preconditions	The user inserted an email that does not correspond to any existing account.	
Minimal Guarantees	None	
Success Guarantees	None	
Trigger	UC5	
Description	**Step**	**Action**
	1	The system alerts the user that has inserted an invalid email.
	2	The sequence of events starts at step 2 of the main flow (UC5).
Technology & data Variations	None	

Alternative flow	Reset password: Invalid identifier	
Release (id)	UC5.2, v0.1, 25/04/2026	
Context of Use	Account management.	
Scope	The system alerts the authenticated user that they have inserted an invalid identifier.	
Level	Subfunction	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to reset password
	System	Manage the user interface, the database and their interactions
Preconditions	The user inserted an identifier that does not correspond to the one created by the system.	
Minimal Guarantees	None	
Success Guarantees	None	
Trigger	UC5	
Description	**Step**	**Action**
	1	The system alerts the user that has inserted an invalid identifier.
	2	The sequence of events starts at step 9 of the main flow (UC5).
Technology & data Variations	None	

Alternative flow	Reset: Invalid password	
Release (id)	UC5.3, v0.1, 25/04/2026	
Context of Use	Account management.	
Scope	The system alerts the anonymous user that they have inserted an invalid password.	
Level	Subfunction	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to reset password
	System	Manage the user interface, the database and their interactions
Preconditions	The user inserted a password that does not satisfy the security requirements or the two passwords do not coincide.	
Minimal Guarantees	No unsecure password is created.	
Success Guarantees	None	
Trigger	UC5	
Description (Main Success Scenario)	**Step**	**Action**
	1	The system alerts the user that has inserted an invalid password, specifying the problem.
	2	The sequence of events starts at step 12 of the main flow (UC5).
Technology & data Variations	None	

Use case	Login	
Release (id)	UC6, v0.1, 20/04/2026	
Context of Use	User authentication.	
Scope	Let an anonymous user login.	
Level	Primary task	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to login
	System	Manage the user interface, the database and their interactions
Preconditions	The user must be logged out.	
Minimal Guarantees	No user login in an account they have no authorization for.	
Success Guarantees	The anonymous user is logged in her account.	
Trigger	The anonymous user selects “login”.	
Description	**Step**	**Action**
	1	The anonymous user selects the functionality “login”.
	2	The system asks the user to insert username and password.
	3	The user inserts the required data.
	4	The system checks whether both the username and password correspond to an existing account.
	5	The system switches to the authenticated user functionalities.
Extension	**Step**	**Other action**
	4a	Invalid username or password: UC5.1

Technology & data Variations	None

Alternative flow	Login: Invalid username or password	
Release (id)	UC6.1, v0.2, 25/04/2026	
Context of Use	User authentication.	
Scope	The system alerts the anonymous user that they have inserted an invalid username or password.	
Level	Subfunction	
Primary actor	Anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user	Wants to login
	System	Manage the user interface, the database and their interactions
Preconditions	The user inserted a pair of username and password that does not correspond to any existing account	
Minimal Guarantees	No user login in an account they have no authorization for.	
Success Guarantees	None	
Trigger	UC5	
Description (Main Success Scenario)	**Step**	**Action**
	1	The system alerts the user that has inserted an invalid username or password, specifying the problem.
	2	The sequence of events starts at step 2 of the main flow (UC5).
Technology & data Variations	None	

Use case	Logout	
Release (id)	UC7, v0.1, 20/04/2026	
Context of Use	User authentication.	
Scope	Let an authenticated user logout.	
Level	Primary task	
Primary actor	Authenticated user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Authenticated user	Wants to logout
	System	Manage the user interface, the database and their interactions
Preconditions	The user must be logged in.	
Minimal Guarantees	None	
Success Guarantees	The authenticated user is logged out.	
Trigger	The anonymous user selects “logout”.	
Description	**Step**	**Action**
	1	The anonymous user selects the functionality “logout”.
	5	The system switches to the anonymous user functionalities.
Extension	**Step**	**Other action**
	-	None
Technology & data Variations	None	

Use case	Ticket display	
Release (id)	UC8, v0.1, 21/04/2026	
Context of Use	Ticketing management.	
Scope	Let a user view the details of a ticket.	
Level	Primary task	
Primary actor	Authenticated user (passenger, train company or network manager) or anonymous user	
Stakeholder & interest	**Stakeholder**	**Interest**
	Authenticated user	Wants to view tickets' details
	Anonymous user	Wants to view tickets' details
	System	Manage the user interface, the database and their interactions
Preconditions	The user must be in an interface where a ticket is displayed. Magari da ultra-specificare gli include Arrivi da: path, train, station	
Minimal Guarantees	None	
Success Guarantees	The user is in an interface displaying the ticket's details.	
Trigger	The user selects "view tickets' details".	
Description	**Step**	**Action**
	1	The authenticated user selects the functionality "view tickets' details".
	2	The system displays the ticket's details.
Extension	**Step**	**Other action**
	-	None
Technology & data Variations	None	

Use case	Ticket purchase	
Release (id)	UC9, v0.2, 25/04/2026	
Context of Use	Ticketing management.	
Scope	Enable a passenger to purchase a ticket.	
Level	Primary task	
Primary actor	Passenger	
Stakeholder & interest	**Stakeholder**	**Interest**
	Passenger	Wants to buy a ticket
	Payment gateway	Manages transaction
	System	Manage the user interface, the database and their interactions
Preconditions	The passenger must be logged in. The passenger must be in an interface displaying the ticket's details.	
Minimal Guarantees	The passenger who cannot pay does not purchase any ticket.	
Success Guarantees	The passenger successfully purchased a ticket.	
Trigger	The passenger selects "Buy ticket".	
Description	**Step**	**Action**
	1	The passenger selects the functionality "Buy ticket".
	2	The system sends a transaction request to the payment gateway.
	3	The payment gateway manages the transaction.
	4	The payment gateway sends the transaction's status to the system.
	5	If the transaction is not successful

	5.1	The system alerts the passenger about the failed transaction.
	6	Otherwise
	6.1	The system alerts the passenger about the completed transaction.
	6.2	The system notifies the passenger of successful purchase with an email.
	6.3	The system saves the ticket in the account archive.
Extension	**Step**	**Other action**
	-	None
Technology & data Variations	None	

Use case	Ticket cancellation	
Release (id)	UC10, v0.1, 21/04/2026	
Context of Use	Ticketing management.	
Scope	Enable a passenger to cancel a purchased ticket.	
Level	Primary task	
Primary actor	Passenger	
Stakeholder & interest	**Stakeholder**	**Interest**
	Passenger	Wants to buy a ticket
	Payment gateway	Manages transaction
	System	Manage the user interface, the database and their interactions
Preconditions	The passenger must be logged in. The passenger must be in an interface displaying the ticket's details. The passenger must have already purchased the ticket.	
Minimal Guarantees	The passenger does not cancel any ticket they have not purchased.	
Success Guarantees	The passenger successfully cancelled a purchased ticket.	
Trigger	The passenger selects "Cancel ticket".	
Description	**Step**	**Action**
	1	The passenger selects the functionality "Cancel ticket".
	2	The system sends a transaction request to the payment gateway.
	3	The payment gateway manages the transaction.
	4	The payment gateway sends the transaction's status to the system.
	5	If the transaction is not successful

	5.1	The system alerts the passenger about the failed transaction.
	6	Otherwise
	6.1	The system alerts the passenger about the completed transaction.
	6.2	The system notifies the passenger of successful purchase with an email.
	6.3	The system saves the ticket's cancellation information in the account archive.
Extension	**Step**	**Other action**
	-	None
Technology & data Variations	None	

Use case	Ticket history	
Release (id)	UC11, v0.1, 21/04/2026	
Context of Use	Ticketing management.	
Scope	Enable a passenger to view her tickets' history, both purchased and cancelled.	
Level	Primary task	
Primary actor	Passenger	
Stakeholder & interest	**Stakeholder**	**Interest**
	Passenger	Wants to view tickets' history
	System	Manage the user interface, the database and their interactions
Preconditions	The passenger must be logged in.	
Minimal Guarantees	None	
Success Guarantees	The passenger successfully view tickets' history.	
Trigger	The passenger selects "Tickets' history".	
Description	**Step**	**Action**
	1	The passenger selects the functionality "Tickets' history".
	2	The system shows the list of tickets purchased and cancelled by that passenger.
Extension	**Step**	**Other action**
	-	None
Technology & data Variations	None	

Use case	Find Path	
Release (id)	UC12, v0.1, 25/04/2026	
Context of Use	Path finding	
Scope	Let an anonymous user or passenger find a path between two stations	
Level	Primary task	
Primary actor	Anonymous user OR Passenger	
Stakeholder & interest	**Stakeholder**	**Interest**
	Anonymous user or Passenger	Find a path
	System	Manage the user interface, the database and their interactions
Preconditions	The user must be either logged out or logged in as a Passenger	
Minimal Guarantees	No impossible path is proposed	
Success Guarantees	The user has a suitable path and can directly select other operations for it.	
Trigger	The user selects “find a path”.	
Description	**Step**	**Action**
	1	The user selects the functionality “find a path”.
	2	The system asks the user to insert a Departing and Arriving station
	3	The user inserts all the requested data.
	4	The system ask the user an Departure or Arrival time
	5	The user provides any of the requested data, possibly confirming zero

	6	The system checks whether at least one path along the mesh between the two stations exists
	7	The system computes whether one or the intersection of multiple service lines (referred to as candidate paths) connects the stations
	8	The system filters out all candidate paths that do not respect Departure and Arrival times, if provided
	9	The system presents the user an option to change path-sorting algorithm from a default option
	10	The system uses the path-sorting algorithm to provide the top-ranking paths computed at step 8
	11	The user selects one path
	12	The system presents to the user all path-specific actions regarding that path (UC 12.1)
Extension	**Step**	**Other action**
	3a	If the user does not insert sufficient data (leaves arrival or departure station blank), display “no arrival/departure station” error and return to step 2
	5a	If the user inserts zero data, default to “Departure” as current time, proceed normally
	6a	If no path is found, produce “Stations disconnected” error and return to step 2
	7a	If no candidate paths are found, produce “No path available” error and return to step 2
	8a	If no candidate paths remain after the filter, display “Too strict Arrival/Departure” error and return to step 4
	9-10a	If the user changes sorting algorithm, the newly selected one is used to repeat from step 8

	9-10b	If two or more paths score equally in selected sorting algorithm, use all other sorting algorithms in a pre-determined order as tie-breakers, and failing that, assign order among them randomly. Proceed as normal
Technology & data Variations	If user is not a logged-in Passenger, ticket actions for the path are displayed but marked as unavailable	

Use case	Purchase Tickets in path	
Release (id)	UC12.1, v0.1, 28/04/2026	
Context of Use	Path finding, Ticket Purchase	
Scope	Let a Passenger directly purchase tickets involved in path	
Level	Subfunction	
Primary actor	Passenger	
Stakeholder & interest	**Stakeholder**	**Interest**
	Passenger	Purchase tickets
	Payment Gateway	Process transaction
	System	Manage the user interface, the database and their interactions
Preconditions	The user must be either logged in as a Passenger	
Minimal Guarantees	No unnecessary tickets are proposed	
Success Guarantees	The user purchased each ticket	
Trigger	The user selects “Tickets” at step 13 of UC12	
Description	**Step**	**Action**
	1	The Passenger selects “Tickets”
	2	The system computes the cheapest set of tickets that cover transit in all trains of the path for the duration of the path itself
	3	All tickets already owned by the passenger are marked as “Owned” on the list.
	4	The system provides such list to the user, along with the option to include(“UC8 Ticket Display”) for each. If all tickets are marked as “Owned”, jump to step 6

	5	Upon UC8 completion (both with purchase or not) in step 4, return to step 3.
	6	The system displays a “All tickets owned!” message Return to UC12 Step 3, marking “Tickets” option as complete
Extension	**Step**	**Other action**
	2a	If no set of tickets exist, produce “Ticketing Impossible” error, return to UC12 step 13, marking “Tickets” option as error
	4a	If a well-formed UC8 option for a ticket cannot be made, and it is not already marked “Owned”, provide “Invalid Ticket” error and mark that option as error. Continue normally
Technology & data Variations	None	

Use case	View Station	
Release (id)	UC13, v0.1, 28/04/2026	
Context of Use	Info Display	
Scope	Let any user view a station	
Level	Primary task	
Primary actor	User	
Stakeholder & interest	**Stakeholder**	**Interest**
	User	View Station
	System	Manage the user interface, the database and their interactions
Preconditions	The station's thumbnail UC 13.1 must be displayed	
Minimal Guarantees	None	
Success Guarantees	The user views the station's and related data	
Trigger	The user selects the station's thumbnail UC13.1	
Description	**Step**	**Action**
	1	The Passenger selects the station's thumbnail UC13.1
	2	The system displays the station's details
	3	The system displays UC 14.1 Train Thumbnail for trains passing through the station
	4	If the user is a Passenger, it displays tickets starting from the station
Extension	**Step**	**Other action**
		None
Technology & data Variations	None	

Use case	Station Thumbnail	
Release (id)	UC13.1, v0.1, 28/04/2026	
Context of Use	Info Display	
Scope	Let any user preview a station's information	
Level	Subfunction	
Primary actor	User	
Stakeholder & interest	**Stakeholder**	**Interest**
	User	See station as result of query
	System	Manage the user interface, the database and their interactions
Preconditions	The station's information is relevant in some other UC	
Minimal Guarantees	The user can select the thumbnail to expand into UC13 View Station	
Success Guarantees	The user selects the thumbnail to transition into UC13 View Station	
Trigger	Return result from some query to the system Embedded into other UC	
Description	**Step**	**Action**
	1	The station information is displayed
	2	The user selects the thumbnail
	3	UC 13 View Station initiates
Extension	**Step**	**Other action**
		None
Technology & data Variations	None	

Use case	View Train	
Release (id)	UC14, v0.1, 28/04/2026	
Context of Use	Info Display	
Scope	Let any user view a train	
Level	Primary Task	
Primary actor	User	
Stakeholder & interest	**Stakeholder**	**Interest**
	User	View Train
	System	Manage the user interface, the database and their interactions
Preconditions	The train's thumbnail UC14.1 must be displayed	
Minimal Guarantees	None	
Success Guarantees	The user views the train's and related data	
Trigger	The user selects the train's thumbnail UC14.1	
Description	**Step**	**Action**
	1	The Passenger selects the train's thumbnail UC14.1
	2	The system displays the train's details
	3	The system displays UC13.1 Station Thumbnail for stations through which the train passes
	4	If the user is a Passenger, it displays tickets for the train
Extension	**Step**	**Other action**
		None
Technology & data Variations	None	

Use case	Train Thumbnail	
Release (id)	UC14.1, v0.1, 28/04/2026	
Context of Use	Info Display	
Scope	Let any user preview train information	
Level	Subfunction	
Primary actor	User	
Stakeholder & interest	**Stakeholder**	**Interest**
	User	See train as result of query
	System	Manage the user interface, the database and their interactions
Preconditions	The train's information is relevant in some other UC	
Minimal Guarantees	The user can select the thumbnail to expand into UC14 View Train	
Success Guarantees	The user selects the thumbnail to transition into UC14 View Train	
Trigger	Return result from some query to the system Embedded into other UC	
Description	**Step**	**Action**
	1	The train's information is displayed
	2	The user selects the thumbnail
	3	UC 14 View Train initiates
Extension	**Step**	**Other action**
		None
Technology & data Variations	None	

Use case	Search Station	
Release (id)	UC15, v0.1, 28/04/2026	
Context of Use	Info Display	
Scope	Let any user find station by name	
Level	Primary Task	
Primary actor	User	
Stakeholder & interest	**Stakeholder**	**Interest**
	User	Find station by query
	System	Manage the user interface, the database and their interactions
Preconditions	None	
Minimal Guarantees	None	
Success Guarantees	The User is given a list of UC13.1 Station Thumbnails of matching stations, or is warned that there are no matches otherwise	
Trigger	User selects “Search Station”	
Description	**Step**	**Action**
	1	The user selects “Search Station”
	2	The system asks the user for the searched station’s name
	3	The user provides a string
	4	The system matches that string agains all known names of stations in database
	5	For all matching stations, an UC13.1 Station Thumbnail is prepared and appended to a return list
	6	The system provides the user the return list

Extension	**Step**	**Other action**
	4a	If no name matches the string, prodice “No match” error and return to step 2
Technology & data Variations	None	

Use case	Search Train	
Release (id)	UC15, v0.1, 28/04/2026	
Context of Use	Info Display	
Scope	Let any user find a train service by ID	
Level	Primary Task	
Primary actor	User	
Stakeholder & interest	**Stakeholder**	**Interest**
	User	Find train line by query
	System	Manage the user interface, the database and their interactions
Preconditions	None	
Minimal Guarantees	None	
Success Guarantees	The User is given a list of UC14.1 Train Thumbnails of matching train lines, or is warned that there are no matches otherwise	
Trigger	User selects “Search Train”	
Description	**Step**	**Action**
	1	The user selects “Search Train”
	2	The system asks the user for the searched train’s identifier
	3	The user provides a string
	4	The system matches that string agains all known identifiers of train lines in database
	5	For all matching trains, an UC14.1 Train Thumbnail is prepared and appended to a return list
	6	The system provides the user the return list

Extension	**Step**	**Other action**
	4a	If no identifier matches the string, prodice “No match” error and return to step 2
Technology & data Variations	None	